

LOCALIZED BOST–CONNES SYSTEMS IN ARITHMETIC TOPOLOGY: LOCAL CLASS FIELD THEORY, LINKING KAKUTANI DICHOTOMIES, AND TOPOLOGICAL PHASE TRANSITIONS

RUQING CHEN

ABSTRACT. We quantize the Mazur–Kapranov–Reznikov dictionary [2, 1, 5, 3] by transporting the localized Bost–Connes theory of [7] to knots in a 3-manifold, and determine which topological data drive the resulting thermodynamics.

Everything rests on one choice. In the dictionary the boundary torus of a knot plays the role of the local field, $H_1(\partial N(K_i); \widehat{\mathbb{Z}}) \leftrightarrow \widehat{K_{\mathfrak{p}}^{\times}} = \pi_{\mathfrak{p}}^{\widehat{\mathbb{Z}}} \times \mathcal{O}_{\mathfrak{p}}^{\times}$, with the meridian corresponding to *inertia* and the longitude to *Frobenius*, so the Bost–Connes parameter must be $\sigma_{K_i} = [\lambda_i]$. The half-lives-half-dies theorem then completes the dictionary: the Lagrangian $\ker(H_1(\partial X) \rightarrow H_1(X))$, spanned for $M = S^3$ by the 0-framed longitudes $\widehat{\lambda}_i = \lambda_i - \sum_j \text{Lk}(K_i, K_j)\mu_j$, corresponds to the principal ideles $\widehat{K_{S,+}^{\times}}$, its intersection with the meridian subgroup to the global units $\widehat{\mathcal{O}_{K,+}^{\times}}$ — which vanishes for $M = S^3$, matching $\mathbb{Z}_+^{\times} = \{1\}$ under $\text{Spec } \mathbb{Z} \leftrightarrow S^3$ — and the subgroup of $H_1(M; \mathbb{Z})$ generated by the knot classes to Cl_S^+ .

With this parameter the KMS_{β} simplex is $\text{Prob}(G_{\mathcal{L}}/\Xi_{\beta}^{\perp})$ with the obstruction group governed by the Gauss linking matrix: a character with finite support F and denominator m satisfies $\chi(\sigma_{K_i}) = \exp(2\pi i \sum_{j \in F} \text{Lk}(K_i, K_j)a_j)$, so its detecting set is $D_{\chi} = \{i : \sum_{j \in F} \text{Lk}(K_i, K_j)b_j \not\equiv 0 \pmod{m}\}$. Two structural facts decide whether D_{χ} is infinite, and we prove both: local finiteness is vacuous for an infinite link in a closed manifold, so components must accumulate; and once they do, a compact Seifert surface can be pierced infinitely often and the linking rows acquire infinite support. We exhibit the simplest such family — infinitely many parallel copies of a knot with framing f — for which D_{χ} omits only F whenever $f \sum_{j \in F} a_j \notin \mathbb{Z}$, so that Ξ_{β} is genuinely β -dependent, although for this family some characters are never detected and the state is never unique. We then prove that every symmetric integer matrix is the linking matrix of a countable tame link in S^3 , and use the matrix of binary digits $\text{Lk}(K_i, K_j) = \text{bit}_{\min(i,j)}(\max(i,j))$ to build a link whose KMS_{β} state is unique for $\beta \leq \beta_c$ and whose simplex is all of $\text{Prob}(\prod_j \widehat{\mathbb{Z}})$ above: the Bost–Connes phase diagram, realized by linking alone, with open or closed broken phase according to the norm. Ambient homology enters twice: the subgroup of $H_1(M; \mathbb{Z})$ generated by the knot classes carries the exact analogue of the $h_S^+ > 1$ obstruction, while the quotient $H_1(M; \mathbb{Z})/\langle [K_i] \rangle$, which the topological group sees and G_S never does, breaks the symmetry at every temperature.

Section 6 isolates why the meridian is the wrong choice: μ_i is a single basis vector of $H_1(S^3 \setminus \mathcal{L})$, so D_{χ} is finite and Ξ_{β} is everything at every temperature. This is what inertia should do — inertia is what an unramified cover kills, and the phase structure lives in the unramified direction. The contrast with $\sigma_{\mathfrak{p}} = (\pi_{\mathfrak{p}})_{q \neq p}$, nontrivial at every other place by the product formula, is exact: reciprocity and linking are two ways of spreading a local generator over infinitely many places.

CONTENTS

1.	The local dictionary and its homological foundation	2
1.1.	Local class field theory for knots	2
1.2.	Half lives, half dies, and the unit group	2
2.	The knot-theoretic Bost–Connes system	3
3.	The linking obstruction group	4

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/VIII-arithmetic-topology/](#).

4. Ambient homological obstructions	6
5. Volume exponents and closed phases	7
6. Contrastive autopsy: why the meridian fails	8
7. The dictionary, and assessment	9
References	10

1. THE LOCAL DICTIONARY AND ITS HOMOLOGICAL FOUNDATION

Let M be a closed, connected, oriented 3-manifold and $\mathcal{L} = \{K_1, K_2, \dots\}$ a countable family of disjoint tame knots. Write $N(\mathcal{L})$ for an open tubular neighbourhood,

$$X := M \setminus N(\mathcal{L}), \quad \partial X = \bigsqcup_i T_i, \quad T_i = \partial N(K_i) \cong T^2,$$

so $H_1(T_i; \mathbb{Z}) = \mathbb{Z}\mu_i \oplus \mathbb{Z}\lambda_i$ is free on the meridian and a chosen longitude. For $M = S^3$ we take λ_i to be the Seifert longitude, and we write $\text{Lk}(K_i, K_j)$ for the linking number when $i \neq j$ and $\text{Lk}(K_i, K_i)$ for the framing of λ_i , so $\text{Lk}(K_i, K_i) = 0$ in S^3 .

1.1. Local class field theory for knots.

Definition 1.1 (The local entry). The dictionary of Mazur, Kapranov–Smirnov, Reznikov and Morishita [2, 1, 5, 3] pairs $\text{Spec } \mathcal{O}_{\mathfrak{p}} \leftrightarrow N(K_i)$ and $\text{Spec } K_{\mathfrak{p}} \leftrightarrow N(K_i) \setminus K_i \simeq T_i$, and the local reciprocity isomorphism $\text{Gal}(K_{\mathfrak{p}}^{ab}/K_{\mathfrak{p}}) \cong \widehat{K_{\mathfrak{p}}^{\times}} = \pi_{\mathfrak{p}}^{\widehat{\mathbb{Z}}} \times \mathcal{O}_{\mathfrak{p}}^{\times}$ [4, Ch. V] with $\pi_1^{ab}(T_i)^{\wedge} = \widehat{\mathbb{Z}}\lambda_i \oplus \widehat{\mathbb{Z}}\mu_i$, the entries being

$$(1.1) \quad \mu_i \longleftrightarrow \text{inertia } \mathcal{O}_{\mathfrak{p}}^{\times}, \quad \lambda_i \longleftrightarrow \text{Frobenius } \pi_{\mathfrak{p}}^{\widehat{\mathbb{Z}}}.$$

This is a convention, not a theorem, and everything below depends on it; the reasons for it are the following. In a cyclic cover of M branched along K_i the meridian carries the local monodromy around the branch locus, and the meridians are exactly what must die in an unramified cover of M , as the inertia subgroup $\mathcal{O}_{\mathfrak{p}}^{\times}$ is exactly what dies in an unramified extension. The unramified quotient $\text{Gal}(K_{\mathfrak{p}}^{ur}/K_{\mathfrak{p}}) = \widehat{\mathbb{Z}}$ is generated by Frobenius and matches $\widehat{\mathbb{Z}}\lambda_i$, the longitude surviving in an unramified cover with its image recording the covering degree along K_i . (The tame quotient of $\text{Gal}(\bar{K}_{\mathfrak{p}}/K_{\mathfrak{p}})$ is $\widehat{\mathbb{Z}}' \rtimes \widehat{\mathbb{Z}}$ and matches $\pi_1(T_i)$ only after abelianization and completion; the matching of the inertia group $\mathcal{O}_{\mathfrak{p}}^{\times}$ with the single generator μ_i is the usual coarsening of the dictionary.)

Remark 1.2 (The choice that decides everything). The informal statement of the dictionary — “primes correspond to knots” — does not say which class on the boundary torus is Frobenius, and the meridians are the visible generators of $H_1(S^3 \setminus \mathcal{L})$ (Alexander duality). Setting $\sigma_{K_i} = [\mu_i]$ is therefore tempting, and yields a theory with no thermodynamics at all; §6 carries out that computation and explains it. Everything below uses $\sigma_{K_i} = [\lambda_i]$.

1.2. Half lives, half dies, and the unit group. Let $\iota_* : H_1(\partial X; \widehat{\mathbb{Z}}) \rightarrow H_1(X; \widehat{\mathbb{Z}})$ be induced by inclusion and write $\mathbb{I}_{\partial} := H_1(\partial X; \widehat{\mathbb{Z}}) = \bigoplus_i (\widehat{\mathbb{Z}}\mu_i \oplus \widehat{\mathbb{Z}}\lambda_i)$.

Theorem 1.3 (The Lagrangian). *For X compact oriented with boundary a disjoint union of n tori,*

$$\dim_{\mathbb{Q}} \ker(H_1(\partial X; \mathbb{Q}) \rightarrow H_1(X; \mathbb{Q})) = \frac{1}{2} \dim_{\mathbb{Q}} H_1(\partial X; \mathbb{Q}) = n,$$

and the kernel is Lagrangian for the intersection form. For $M = S^3$ it is spanned by the 0-framed longitudes

$$(1.2) \quad \widetilde{\lambda}_i := \lambda_i - \sum_j \text{Lk}(K_i, K_j) \mu_j.$$

Proof. The first statement is the classical half-lives-half-dies lemma, from the long exact sequence of $(X, \partial X)$ with Poincaré–Lefschetz duality [6, Ch. 4]. For $M = S^3$, Alexander duality gives $H_1(X) = \bigoplus_j \mathbb{Z}\mu_j$ with $\iota_*\mu_i = \mu_i$ and $\iota_*\lambda_i = \sum_j \text{Lk}(K_i, K_j)\mu_j$, so $\sum_i (p_i\mu_i + q_i\lambda_i)$ lies in the kernel precisely when $p_j = -\sum_i q_i \text{Lk}(K_i, K_j)$ for every j ; this is free of rank n on the q_i , with basis (1.2). $\square$

Theorem 1.4 (The completed dictionary). *Let $\mathcal{L}$ be finite with n components, $\iota_* : \mathbb{I}_\partial \rightarrow H_1(X; \widehat{\mathbb{Z}})$, and let $\nu : H_1(X; \widehat{\mathbb{Z}}) \rightarrow H_1(M; \widehat{\mathbb{Z}})$ be induced by inclusion, so that $\nu\iota_*\mu_i = 0$ and $\nu\iota_*\lambda_i = [K_i]$. Put*

$$G_{\mathcal{L}}^{\text{idl}} := \mathbb{I}_\partial / \ker \iota_* = \iota_*(\mathbb{I}_\partial) \leq H_1(X; \widehat{\mathbb{Z}}), \quad \text{Cl}_{\mathcal{L}} := \langle [K_i] : i \rangle \leq H_1(M; \widehat{\mathbb{Z}}).$$

Then the idele exact sequence

$$1 \longrightarrow U_S / \overline{\mathcal{O}_{K,+}^\times} \longrightarrow G_S = \mathbb{I}_S / \overline{K_{S,+}^\times} \longrightarrow \text{Cl}_S^+ \longrightarrow 1$$

of [7, Prop. 2.5], in which $\text{Cl}_S^+ \leq \text{Cl}_K^+$ is the subgroup generated by the classes of the primes of S and the last map is the divisor map, corresponds term by term to the exact sequence

$$1 \longrightarrow \left(\bigoplus_i \widehat{\mathbb{Z}}\mu_i \right) / \left(\ker \iota_* \cap \bigoplus_i \widehat{\mathbb{Z}}\mu_i \right) \longrightarrow G_{\mathcal{L}}^{\text{idl}} \xrightarrow{\nu} \text{Cl}_{\mathcal{L}} \longrightarrow 1,$$

via

$$\mathbb{I}_S \leftrightarrow \mathbb{I}_\partial, \quad U_S \leftrightarrow \bigoplus_i \widehat{\mathbb{Z}}\mu_i, \quad \overline{K_{S,+}^\times} \leftrightarrow \ker \iota_*, \quad \overline{\mathcal{O}_{K,+}^\times} \leftrightarrow \ker \iota_* \cap \bigoplus_i \widehat{\mathbb{Z}}\mu_i, \quad \text{Cl}_S^+ \leftrightarrow \text{Cl}_{\mathcal{L}}.$$

For $M = S^3$ the unit term and $\text{Cl}_{\mathcal{L}}$ both vanish, matching $\mathbb{Z}_+^\times = \{1\}$ and $\text{Cl}(\mathbb{Z}) = 1$ under $\text{Spec } \mathbb{Z} \leftrightarrow S^3$, and $G_{\mathcal{L}}^{\text{idl}} = H_1(X; \widehat{\mathbb{Z}}) = \bigoplus_i \widehat{\mathbb{Z}}\mu_i$. In general $H_1(X; \widehat{\mathbb{Z}})/G_{\mathcal{L}}^{\text{idl}} \cong H_1(M; \widehat{\mathbb{Z}})/\text{Cl}_{\mathcal{L}}$, a quotient with no arithmetic counterpart.

Proof. The first two assignments are Definition 1.1, $K_p^\times$ decomposing into a valuation part and a unit part matching $\widehat{\mathbb{Z}}\lambda_i$ and $\widehat{\mathbb{Z}}\mu_i$. The principal ideles are those dying in the global quotient, matching $\ker \iota_*$; the global units are the principal ideles that are local units at every place, matching the intersection with the meridian subgroup; and the divisor map, which reads off the valuations of an idele, matches ν , which reads off the longitude coefficients since $\nu\iota_*\mu_i = 0$ and $\nu\iota_*\lambda_i = [K_i]$, the longitude being isotopic to the core of $N(K_i)$. Exactness: ν maps $G_{\mathcal{L}}^{\text{idl}}$ onto the subgroup generated by the $[K_i]$; its kernel is $\iota_*^{-1}(\ker \nu)/\ker \iota_*$, and $\ker \nu = \langle \iota_*\mu_i \rangle$ because M is obtained from X by Dehn filling along the meridians, so that $H_1(M) = H_1(X)/\langle \mu_i \rangle$; if $\iota_*\xi = \iota_*\eta$ with $\eta \in \bigoplus \widehat{\mathbb{Z}}\mu_i$ then $\xi \equiv \eta$ modulo $\ker \iota_*$, which identifies the kernel with the image of $\bigoplus \widehat{\mathbb{Z}}\mu_i$. For $M = S^3$, by (1.2) an element of $\ker \iota_*$ is $\sum_i q_i \tilde{\lambda}_i$, which lies in $\bigoplus \widehat{\mathbb{Z}}\mu_i$ only if every $q_i = 0$, and $H_1(S^3) = 0$. The last assertion is $H_1(X)/\langle \iota_*\mu_i, \iota_*\lambda_i \rangle = (H_1(X)/\langle \mu_i \rangle)/\langle [K_i] \rangle$. $\square$

Remark 1.5 (Numerical check). For random symmetric linking matrices with zero diagonal and $n \in \{2, 3, 5, 8\}$, the rank of $\ker \iota_*$ was computed to be exactly $n = \frac{1}{2} \dim H_1(\partial X)$ in every case, and $\ker \iota_* \cap \bigoplus \widehat{\mathbb{Z}}\mu_i = 0$ in every case.

2. THE KNOT-THEORETIC BOST–CONNES SYSTEM

Remark 2.1 (Infinite links must accumulate). For infinite $\mathcal{L}$ in a closed M one cannot require that every compact set meet only finitely many components: taking the compact set to be M would force $\mathcal{L}$ finite. The components of an infinite link therefore accumulate on a nonempty closed set. This is not a defect; by Theorem 3.3 it is exactly what allows the linking rows to have infinite support, and hence what makes the theory nonempty.

Definition 2.2 (Norm). A norm is a function $N : \mathcal{L} \rightarrow [2, \infty)$, extended to $J_{\mathcal{L}} = \bigoplus_i \mathbb{N} \cdot K_i$ by $N(D) = \prod_i N(K_i)^{d_i}$; this is well defined and multiplicative because $J_{\mathcal{L}}$ is free. Candidates on generators: $N(K_i) = e^{\text{Vol}(M \setminus K_i)}$ for hyperbolic components, $e^{2\pi \deg \Delta_{K_i}}$, or $e^{c \ell(\mu_i)}$.

Remark 2.3 (Geometric only on generators). Multiplicativity is imposed. For $D = K_1 + K_2$ the value $N(K_1)N(K_2)$ is not $e^{\text{Vol}(M \setminus (K_1 \cup K_2))}$, hyperbolic volume being neither additive over disjoint unions nor monotone in the naive direction. The geometric reading survives on generators only; N-multiplicativity, a theorem about ideals, is here a definition.

Definition 2.4 (The system). Let

$$G_{\mathcal{L}} := \varprojlim_{F,n} H_1(M \setminus \mathcal{L}_F; \mathbb{Z}/n\mathbb{Z}), \quad \sigma_{K_i} := [\lambda_i] \in G_{\mathcal{L}},$$

the limit over finite sublinks F and integers n , the transition maps being induced by the inclusions $M \setminus \mathcal{L}_{F'} \subset M \setminus \mathcal{L}_F$ for $F \subset F'$; $G_{\mathcal{L}}$ is a compact abelian group, and $[\lambda_i]$ is the compatible family of classes of the longitude of K_i in the complements of the finite sublinks containing K_i . For finite $\mathcal{L}$, $G_{\mathcal{L}} = H_1(X; \widehat{\mathbb{Z}})$. Let $X_{\mathcal{L}} = \overline{\mathbb{N}}^{\mathcal{L}}$ with $\overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$, on which $J_{\mathcal{L}}$ acts by $v \mapsto v + D$, and put

$$Y_{\mathcal{L}} := X_{\mathcal{L}} \times G_{\mathcal{L}}, \quad D \cdot (v, g) = (v + D, g + \sum_i d_i \sigma_{K_i}),$$

$$\mathcal{A}_{M,\mathcal{L}} := C(Y_{\mathcal{L}}) \rtimes J_{\mathcal{L}}, \quad \sigma_t(\mu_D) = N(D)^{it} \mu_D,$$

the semigroup crossed product being the unital full corner of the groupoid algebra of the dilated action of $I_{\mathcal{L}} = \mathbb{Z}^{(\mathcal{L})}$, exactly as $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ in [7, §2]; measure-theoretically $Y_S = \widehat{\mathcal{O}}_S \times_{U_S} G_S$ is $\mathbb{N}^S \times G_S$ with the same action, by [7, Lem. 2.10].

Proposition 2.5 (No new operator algebra is needed). $\mathcal{A}_{M,\mathcal{L}}$ is the system of [7] attached to the quadruple $(\mathcal{L}, J_{\mathcal{L}}, G_{\mathcal{L}}, N)$ with σ_{K_i} in place of σ_p : it is the abelian case of the Galois-twisted groupoid of Paper VII of the series with $\text{Frob}_{K_i} = [\lambda_i]$. The classification theorems [7, Thm. 3.6, Thm. 3.9] and the factor-type results of [8] that depend only on the sequence $(N(K_i), \chi(\sigma_{K_i}))$ apply verbatim. Results of the series whose proofs use arithmetic input — Chebotarev density, prime number theorems, class numbers, the finiteness of the unit group — do not transfer automatically and have to be re-proved, or fail, in the topological setting.

Proof. The proofs of [7, Thm. 3.6, Thm. 3.9] use only that J_S is a free abelian monoid on a countable set acting on $\overline{\mathbb{N}}^S \times G_S$ as above, that G_S is compact abelian with prescribed elements σ_p , that N is multiplicative with values in $[2, \infty)$, and the Kolmogorov zero-one law for the forced product measure; they do not use that the σ_p generate a dense subgroup, since Ξ_{β} is defined by characters of the whole group. The quadruple $(\mathcal{L}, J_{\mathcal{L}}, G_{\mathcal{L}}, N)$ satisfies all of this. No property of prime ideals is used in the classification; arithmetic enters only when the criterion is evaluated. $\square$

3. THE LINKING OBSTRUCTION GROUP

Theorem 3.1 (Topological KMS classification). The KMS_{β} simplex of $\mathcal{A}_{M,\mathcal{L}}$ is affinely isomorphic to $\text{Prob}(G_{\mathcal{L}}/\Xi_{\beta}(M, \mathcal{L})^{\perp})$, where

$$(3.1) \quad \Xi_{\beta}(M, \mathcal{L}) = \left\{ \chi \in \widehat{G_{\mathcal{L}}} : \sum_i N(K_i)^{-\beta} |1 - \chi([\lambda_i])|^2 < \infty \right\},$$

and the state is unique if and only if every nontrivial χ has $\sum_{i \in D_{\chi}} N(K_i)^{-\beta} = \infty$, where $D_{\chi} = \{i : \chi([\lambda_i]) \neq 1\}$.

Proof. Proposition 2.5 together with [7, Thm. 3.6, Thm. 3.9]. Note that a character trivial on every σ_{K_i} has $D_{\chi} = \emptyset$ and lies in Ξ_{β} for all β ; the criterion therefore fails, at every temperature, unless the σ_{K_i} generate a dense subgroup of $G_{\mathcal{L}}$. In the arithmetic case this density is a form of Chebotarev's theorem; here it is a condition on the linking matrix. $\square$

Theorem 3.2 (The Gauss linking matrix formula). *Let $M = S^3$, so $\widehat{G}_{\mathcal{L}} = \bigoplus_j \mathbb{Q}/\mathbb{Z}$ by Alexander duality. For $\chi = (a_j)$ with finite support F and common denominator m , writing $b_j = ma_j$,*

$$(3.2) \quad \chi(\sigma_{K_i}) = \exp\left(2\pi i \sum_{j \in F} \text{Lk}(K_i, K_j) a_j\right), \quad D_\chi = \left\{i : \sum_{j \in F} \text{Lk}(K_i, K_j) b_j \not\equiv 0 \pmod{m}\right\}.$$

For χ of order m the weight $|1 - \chi([\lambda_i])|^2$ is bounded above and below by positive constants on D_χ , so

$$\chi \in \Xi_\beta(S^3, \mathcal{L}) \iff \sum_{i \in D_\chi} N(K_i)^{-\beta} < \infty.$$

Proof. For $M = S^3$ and each finite sublink, $H_1(M \setminus \mathcal{L}_F; \mathbb{Z})$ is free on the meridians μ_j , $j \in F$, by Alexander duality, the transition maps kill the meridians of the omitted components, and $G_{\mathcal{L}} = \prod_j \widehat{\mathbb{Z}}\mu_j$, whose dual is the direct sum of the duals of the factors. Then $\iota_*\lambda_i = \sum_{j \in F} \text{Lk}(K_i, K_j)\mu_j$ in each finite complement, by Theorem 1.3 with our framing convention; evaluate χ . The weight takes finitely many values, all nonzero on D_χ , since χ has order m . $\square$

Theorem 3.3 (Infinite detecting sets exist). *Let $K \subset S^3$ be a knot, $f \in \mathbb{Z}$ a framing, and let $\mathcal{L} = \{K_1, K_2, \dots\}$ consist of infinitely many parallel copies of K with framing f , accumulating on K , so that $\text{Lk}(K_i, K_j) = f$ for all $i \neq j$. Let $\chi = (a_j)_{j \in F}$ and $\Sigma_F = \sum_{j \in F} a_j$. If $f\Sigma_F \notin \mathbb{Z}$ then*

$$D_\chi \supseteq \{i : i \notin F\},$$

so D_χ is cofinite and $\chi \in \Xi_\beta(S^3, \mathcal{L})$ if and only if $\sum_i N(K_i)^{-\beta} < \infty$. Consequently Ξ_β is genuinely β -dependent and changes at the convergence exponent of the norm sequence. If $f\Sigma_F \in \mathbb{Z}$, however, $D_\chi \subseteq F$ is finite and $\chi \in \Xi_\beta$ for every β ; such characters exist for every f (take $F = \{1, 2\}$, $a_1 = a_2 = \frac{1}{2}$), so for this family the KMS_β state is never unique.

Proof. For $i \notin F$ we have $\text{Lk}(K_i, K_j) = f$ for every $j \in F$, so by (3.2) $\chi(\sigma_{K_i}) = \exp(2\pi i f\Sigma_F)$, which is $\neq 1$ iff $f\Sigma_F \notin \mathbb{Z}$. The rest is Theorem 3.2. $\square$

Lemma 3.4 (Every symmetric matrix is a linking matrix). *Let $(\ell_{ij})_{i \neq j}$ be any symmetric family of integers indexed by pairs of distinct positive integers. There is a countable family $\mathcal{L} = \{K_1, K_2, \dots\}$ of pairwise disjoint tame knots in S^3 , each individually unknotted, with $\text{Lk}(K_i, K_j) = \ell_{ij}$ for all $i \neq j$. If some column $(\ell_{ij})_i$ has infinite support, the family accumulates on K_j .*

Proof. Construct K_n inductively, K_1 an unknot. Given the disjoint tame knots $K_1, \dots, K_{n-1}$, take for each $j < n$ with $\ell_{nj} \neq 0$ a collection of $|\ell_{nj}|$ small meridian circles of K_j , oriented so that each links K_j with linking number $\text{sgn } \ell_{nj}$, chosen of radius less than 2^{-n} and disjoint from everything constructed so far; if all $\ell_{nj} = 0$ take one small unknotted circle far from the previous components. Join these circles by bands along disjoint arcs in the complement of $K_1 \cup \dots \cup K_{n-1}$ and of the circles, orienting the bands compatibly; the band-connected sum of finitely many disjoint oriented circles is a single tame knot K_n , unknotted since the circles bound disjoint disks which the bands can be chosen to avoid, and it is homologous in $S^3 \setminus K_j$ to the sum of the circles, so $\text{Lk}(K_n, K_j) = \ell_{nj}$ for every $j < n$; by symmetry of the linking number this fixes $\text{Lk}(K_j, K_n)$ as well. The components are pairwise disjoint by construction, and if column j is infinitely supported, infinitely many K_n pass within 2^{-n} of K_j . $\square$

Theorem 3.5 (A link with the Bost–Connes phase diagram). *Let $\mathcal{L}$ be a link in S^3 realizing, by Lemma 3.4, the linking numbers*

$$\text{Lk}(K_i, K_j) = \text{bit}_{\min(i,j)}(\max(i,j)) \in \{0, 1\} \quad (i \neq j),$$

where $\text{bit}_k(n)$ is the coefficient of 2^k in the binary expansion of n , and let N be a norm with $N(K_i) = (i+1)^\gamma$ for some $\gamma > 0$, so that $\sum_i N(K_i)^{-\beta}$ converges iff $\beta > \beta_c := 1/\gamma$. Then

$$\Xi_\beta(S^3, \mathcal{L}) = \{1\} \text{ for } \beta \leq \beta_c, \quad \Xi_\beta(S^3, \mathcal{L}) = \widehat{G}_{\mathcal{L}} \text{ for } \beta > \beta_c :$$

the KMS_β state of $\mathcal{A}_{S^3, \mathcal{L}}$ is unique for $\beta \leq \beta_c$, and for $\beta > \beta_c$ the simplex is $\text{Prob}(G_{\mathcal{L}}) = \text{Prob}(\prod_j \widehat{\mathbb{Z}})$, on which $G_{\mathcal{L}}$ acts freely and transitively on the extreme points. Every column of the linking matrix is infinitely supported, so $\mathcal{L}$ accumulates on every one of its components.

Proof. For $\beta > \beta_c$ every sum in (3.1) converges, so $\Xi_\beta = \widehat{G_{\mathcal{L}}}$ and $\Xi_\beta^\perp = 0$. Let $\beta \leq \beta_c$ and let $\chi \neq 1$ have support F , denominator $m \geq 2$ and $b = (b_j)_{j \in F}$ not all $\equiv 0 \pmod{m}$. For $i > \max F$, (3.2) gives $i \in D_\chi$ iff $\sum_{j \in F} \text{bit}_j(i) b_j \not\equiv 0 \pmod{m}$. Put $L = \max F + 1$. On any block $[k2^L, (k+1)2^L)$ of 2^L consecutive integers the residue $i \bmod 2^L$ runs over all values once, so the vector $(\text{bit}_j(i))_{j \in F} \in \{0, 1\}^F$ takes each value exactly $2^{L-|F|}$ times. Fix $j_0 \in F$ with $b_{j_0} \not\equiv 0$ and pair each $\varepsilon \in \{0, 1\}^F$ with the vector ε' differing from it in the j_0 -th coordinate only: $\sum_j \varepsilon_j b_j$ and $\sum_j \varepsilon'_j b_j$ differ by $\pm b_{j_0} \not\equiv 0$, so at most one of them is $\equiv 0$. Hence every block contains at least 2^{L-1} elements of D_χ , apart from the finitely many $i \leq \max F$, and D_χ has lower asymptotic density at least $\frac{1}{2}$. A set D of positive lower density c has k -th element at most k/c for k large, hence $\sum_{i \in D} (i+1)^{-s} \geq \sum_k (k/c+1)^{-s} = \infty$ for every $s \leq 1$, and $\gamma\beta \leq 1$; so $\chi \notin \Xi_\beta$. Thus $\Xi_\beta = \{1\}$ and the state is unique by Theorem 3.1. Column j of the matrix contains $\text{bit}_j(i)$ for all $i > j$, which is 1 for infinitely many i . $\square$

Remark 3.6. Theorem 3.5 is the topological Bost–Connes phase diagram in its original shape: below and at the critical temperature a unique equilibrium state, above it spontaneous symmetry breaking with the full compact group of symmetries acting simply transitively on the extreme states, and no intermediate regime. Intermediate chains $\{\Xi_\beta\}$, as realized arithmetically in [8, Thm. 5.1] by congruence conditions on primes, can be realized by prescribing rows of the linking matrix whose congruence classes are controlled; we do not pursue this. What Theorem 3.5 cannot control is the norm: the norm is prescribed here, and Section 5 records what happens when it is asked to be geometric.

Remark 3.7 (Why local finiteness had to go). If $\mathcal{L}$ were locally finite, a compact Seifert surface Σ_j for K_j would meet only finitely many components, so $\text{Lk}(K_j, K_i) \neq 0$ for finitely many i ; by symmetry of the linking number every column of the matrix would then be finitely supported and every D_χ finite, returning the empty theory of §6. By Remark 2.1 local finiteness is in any case impossible for infinite $\mathcal{L}$ in closed M . Accumulation is what lets Σ_j be pierced infinitely often, and Theorem 3.3 realizes this in the simplest way; Lemma 3.4 shows that nothing beyond symmetry constrains the linking matrix of a countable tame link.

Remark 3.8 (Numerical illustration). For random integer linking data $\text{Lk}(K_i, K_j) \in [-4, 4]$ over 400 components and a character supported on two meridians, the detecting set contained 45–47% of components for $m = 2$, 65–69% for $m = 3$ and 78–90% for $m = 5$. This illustrates the generic behaviour of the congruence condition in (3.2); it is not itself a link realization, the random matrices being constrained neither to symmetry nor to arising from an embedded family. The rigorous statements are Theorems 3.3 and 3.5.

4. AMBIENT HOMOLOGICAL OBSTRUCTIONS

Let $\nu : G_{\mathcal{L}} \rightarrow H_1(M; \widehat{\mathbb{Z}})$ be the map induced by the inclusions $M \setminus \mathcal{L}_F \subset M$; it is surjective, since $H_1(M \setminus \mathcal{L}_F) \rightarrow H_1(M)$ is onto for every finite F and the groups are compact, and $\nu(\sigma_{K_i}) = [K_i]$, $\nu([\mu_i]) = 0$. As in Theorem 1.4 put $\text{Cl}_{\mathcal{L}} = \langle [K_i] : i \rangle \leq H_1(M; \mathbb{Z})$.

Theorem 4.1 (Ambient homology). *Let ψ be a character of $H_1(M; \widehat{\mathbb{Z}})$ and $\chi = \psi \circ \nu \in \widehat{G_{\mathcal{L}}}$. Then $\chi(\sigma_{K_i}) = \psi([K_i])$ for every i , and*

$$\chi \in \Xi_\beta(M, \mathcal{L}) \iff \sum_{i: \psi([K_i]) \neq 1} N(K_i)^{-\beta} < \infty.$$

Consequently:

- (1) If ψ is trivial on $\text{Cl}_{\mathcal{L}}$, then $\chi \in \Xi_{\beta}(M, \mathcal{L})$ for every $\beta > 0$. The characters of $H_1(M; \widehat{\mathbb{Z}})/\overline{\text{Cl}_{\mathcal{L}}}$ pull back injectively to Ξ_{β} , so with $h_{\mathcal{L}}^0 := |H_1(M; \mathbb{Z})/\text{Cl}_{\mathcal{L}}|$ the KMS_{β} simplex has at least $h_{\mathcal{L}}^0$ extreme points at every temperature, and uniqueness never holds unless the knot classes generate $H_1(M; \mathbb{Z})$. This obstruction has no arithmetic counterpart: G_S maps onto the subgroup Cl_S^+ generated by the primes of S and never sees $\text{Cl}_K^+/\text{Cl}_S^+$.
- (2) If ψ is nontrivial on $\text{Cl}_{\mathcal{L}}$ and the knots whose classes lie outside $\ker \psi \cap \text{Cl}_{\mathcal{L}}$ form a β -summable set, then $\chi \in \Xi_{\beta}(M, \mathcal{L})$ and the KMS_{β} state is not unique. This is the exact counterpart of the unramified class-group obstruction [7, Prop. 4.1]: a proper subgroup $H < \text{Cl}_{\mathcal{L}}$ containing the classes of all but a β -summable set of components, with $\text{Cl}_{\mathcal{L}}$ in place of Cl_S^+ and $[K_i]$ in place of $[\mathfrak{p}]$.

Proof. $\chi(\sigma_{K_i}) = \psi(\nu[\lambda_i]) = \psi([K_i])$. The group $H_1(M; \widehat{\mathbb{Z}})$ is profinite, so ψ has finite order and $|1 - \psi([K_i])|^2$ takes finitely many values, bounded below by a positive constant where it is nonzero; the displayed equivalence follows from (3.1). (1): all summands vanish. Distinct characters of $H_1(M; \widehat{\mathbb{Z}})$ pull back to distinct characters of $G_{\mathcal{L}}$ since ν is surjective, and a group of $h_{\mathcal{L}}^0$ characters inside Ξ_{β} forces $|G_{\mathcal{L}}/\Xi_{\beta}^{\perp}| \geq h_{\mathcal{L}}^0$. (2): the sum in the equivalence runs over the components with $[K_i] \notin \ker \psi$. $\square$

Corollary 4.2 (The parallel, corrected). *The counterpart of the unramified obstruction of [7, §4.1] is Theorem 4.1(2), under*

$$\text{Cl}_S^+ = \langle [\mathfrak{p}] : \mathfrak{p} \in S \rangle \leq \text{Cl}_K^+ \longleftrightarrow \text{Cl}_{\mathcal{L}} = \langle [K_i] : i \rangle \leq H_1(M; \mathbb{Z}), \quad h_S^+ = |\text{Cl}_S^+| \leftrightarrow |\text{Cl}_{\mathcal{L}}|.$$

The quotient $H_1(M; \mathbb{Z})/\text{Cl}_{\mathcal{L}}$ of Theorem 4.1(1) is a purely topological addition: the complement of a link remembers the homology of the ambient manifold, while the S -ideles remember nothing of the class group beyond the classes of S .

Example 4.3. Let $M = L(p, 1)$, $H_1(M; \mathbb{Z}) = \mathbb{Z}/p$. If all K_i are null-homologous, $\text{Cl}_{\mathcal{L}} = 0$, $h_{\mathcal{L}}^0 = p$, and the simplex has at least p extreme points at every β by Theorem 4.1(1); the h_S^+ -type obstruction is absent, as for a field with $\text{Cl}_S^+ = 1$. If instead some components generate $H_1(M)$, then $h_{\mathcal{L}}^0 = 1$, $\text{Cl}_{\mathcal{L}} = \mathbb{Z}/p$, and Theorem 4.1(2) applies: for a character ψ of $\mathbb{Z}/p$, the pull-back lies in Ξ_{β} iff the components with $\psi([K_i]) \neq 1$ are β -summable, so a link in which all but a summable family of components are null-homologous has broken symmetry at every β , exactly as an S whose primes lie, up to a summable set, in a proper subgroup of Cl_S^+ . What matters is the distribution of the homology classes of the knots, as it is the distribution of the ideal classes of the primes. For $M = S^3$ both obstructions are absent, and by Theorems 3.3 and 3.5 the whole phase structure comes from linking.

5. VOLUME EXPONENTS AND CLOSED PHASES

Definition 5.1. For hyperbolic $\mathcal{L}$ set $\beta_c(\mathcal{L}) = \inf\{\beta > 0 : \sum_i e^{-\beta \text{Vol}(M \setminus K_i)} < \infty\}$.

Proposition 5.2 (The full family is too large). *Over the family of all hyperbolic knots in S^3 the series $\sum_i e^{-\beta \text{Vol}_i}$ diverges for every $\beta > 0$, so $\beta_c = \infty$, and a finite volume exponent requires a sparse subfamily.*

Proof. By Thurston's hyperbolic Dehn surgery theorem [6, Ch. 5] the twist knots K_n , obtained by $1/n$ -surgery on one component of the Whitehead link, are hyperbolic for n large with volumes increasing to the volume $V_W \approx 3.66$ of the Whitehead link complement. So infinitely many hyperbolic knots have volume below V_W , and the series has infinitely many terms bounded below by $e^{-\beta V_W}$. $\square$

Remark 5.3. The count of hyperbolic 3-manifolds of volume at most V is not finite in dimension 3, precisely because of Dehn filling; the growth rate V^{cV} of Burger–Gelder–Lubotzky–Mozes concerns dimension ≥ 4 . The divergence in Proposition 5.2 is therefore stronger than a growth estimate: the volume spectrum of knots has accumulation points, and the volume series diverges term by term near each of them.

Proposition 5.4 (Open and closed broken phases). *Let $\mathcal{L}$ be the link of Theorem 3.5 and let N be any norm with $\sum_{i \in D} N(K_i)^{-\beta} = \infty$ for every $\beta < \beta_c$ and every set D of positive lower density, where β_c is the convergence exponent of $\sum_i N(K_i)^{-\beta}$. Then the KMS_β state is unique for $\beta < \beta_c$ and the simplex is $\text{Prob}(G_{\mathcal{L}})$ for $\beta > \beta_c$, and at $\beta = \beta_c$ the state is unique if $\sum_i N(K_i)^{-\beta_c} = \infty$ and the simplex is $\text{Prob}(G_{\mathcal{L}})$ if $\sum_i N(K_i)^{-\beta_c} < \infty$: the broken phase is the open half-line (β_c, ∞) for $N(K_i) = (i+1)^{1/\beta_c}$ and the closed half-line $[\beta_c, \infty)$ for $N(K_i) = ((i+1)\log^2(i+2))^{1/\beta_c}$, exactly as in [7, Thm. 5.2].*

Proof. The proof of Theorem 3.5 shows that every nontrivial character is detected on a set of lower density at least $\frac{1}{2}$, so for $\beta < \beta_c$ the hypothesis on N gives $\Xi_\beta = \{1\}$; for $\beta > \beta_c$ all sums converge. At β_c : if the full series diverges, so does every sum over a set D of positive lower density c for the two norms named, since their terms a_i are eventually decreasing and the k -th element of D is at most k/c , so that $\sum_{i \in D} a_i \geq \sum_k a_{\lceil k/c \rceil} \geq \int_1^\infty a(x/c) dx = c \int_{1/c}^\infty a = \infty$; if it converges, all subsums converge. The two norms have $\sum_i (i+1)^{-1} = \infty$ and $\sum_i ((i+1)\log^2(i+2))^{-1} < \infty$, and for $\beta < \beta_c$ their β -sums over positive-density sets diverge by partial summation. $\square$

Remark 5.5. With a prescribed norm the theory is complete: Theorem 3.5 and Proposition 5.4 realize both the open and the closed phase diagram of [7, Thm. 5.2] by linking alone. What we have not exhibited is a family that is simultaneously linking-rich in the sense of Theorem 3.5 and *geometric* in its norm: the components of Lemma 3.4 are unknots, and controlling the hyperbolic volumes of an accumulating family — whose complements degenerate — is the concrete gap on the geometric side. Definition 2.2 allows any norm, but the dictionary’s promise that N_p corresponds to a geometric quantity is kept only on generators, by Remark 2.3, and not yet by any accumulating family.

6. CONTRASTIVE AUTOPSY: WHY THE MERIDIAN FAILS

Theorem 6.1 (The inertia model has no thermodynamics). *Let $M = S^3$ and define the system with $\sigma_{K_i} = [\mu_i]$ instead. Then every $\chi \in \widehat{G}_{\mathcal{L}}$ has finite detecting set, $\Xi_\beta = \widehat{G}_{\mathcal{L}}$ for every $\beta > 0$, the KMS_β state is unique at every temperature, and no phase transition occurs — for any link, any norm, any β .*

Proof. As in the proof of Theorem 3.2, $G_{\mathcal{L}} \cong \prod_i \widehat{\mathbb{Z}}\mu_i$ and $\widehat{G}_{\mathcal{L}} = \bigoplus_i \mathbb{Q}/\mathbb{Z}$. A character is thus nontrivial in finitely many coordinates, and since μ_i is the i -th basis vector, D_χ is that finite set. $\square$

Proposition 6.2 (The arithmetic contrast). *Let $K = \mathbb{Q}$. In $G_S = \mathbb{I}_S/\overline{\mathbb{Q}_{S,+}^\times}$, with the convention $\sigma_a = r(s_a)^{-1}$ of [7, (2.2)],*

$$(6.1) \quad \sigma_p = \theta_p = (p)_{\ell \neq p} \times (1)_{\ell=p} \in U_S = \prod_{\ell} \mathbb{Z}_{\ell}^{\times},$$

nontrivial at every place $\ell \neq p$. Consequently a character supported at a single place ℓ_0 , with component χ_{ℓ_0} a character of $\mathbb{Z}_{\ell_0}^{\times}$ of finite order, has $\chi(\sigma_p) = \chi_{\ell_0}(p) = \chi(\text{Frob}_p)^{-1}$ for $p \neq \ell_0$, with detecting set of positive Dirichlet density.

Proof. σ_p is the class of the inverse of the idele s_p with p at p and 1 elsewhere; since p lies in $\mathbb{Q}_{S,+}^\times$ the diagonal idele $(p, p, \dots)$ is trivial in G_S , and $p^{-1}s_p$ has component p^{-1} at $\ell \neq p$ and 1 at p , so its inverse is (6.1); this is [7, Lem. 2.10(1)]. Evaluating χ at (6.1) gives $\chi_{\ell_0}(p)$, and $\chi(\sigma_p) = \chi(\text{Frob}_p)^{-1}$ is [7, Lem. 3.8]. A nontrivial χ_{ℓ_0} factors through $(\mathbb{Z}/\ell_0^k)^\times$, so D_χ is a union of arithmetic progressions, of positive Dirichlet density by Dirichlet’s theorem [4, Ch. VII]. $\square$

Remark 6.3 (The moral). The two computations are the same statement. Arithmetically it is the product formula that spreads a one-place idele across all others; topologically it is the relation $\lambda_i = \sum_j \text{Lk}(K_i, K_j)\mu_j$ that spreads the longitude across all linked components. *Inertia is local and carries no thermodynamics; Frobenius is global and carries the linking matrix.* The meridian model of Theorem 6.1 is not so much a failed construction as a correct computation of the wrong object: it says inertia classes are thermodynamically inert, which is what inertia is for. Numerically, for $\ell_0 = 5, 7, 11$ with χ_{ℓ_0} the Legendre symbol the arithmetic detecting sets contained 51.5%, 52.8% and 52.2% of the primes below 2000, matching the topological densities of Remark 3.8.

7. THE DICTIONARY, AND ASSESSMENT

Arithmetic [7]	Knot topology	Reference
$\mathfrak{p} \in S$	$K_i \in \mathcal{L}$	—
$K_{\mathfrak{p}}^\times$	$H_1(T_i; \widehat{\mathbb{Z}})$	Lem. 1.1
$\mathcal{O}_{\mathfrak{p}}^\times$ (inertia)	μ_i (meridian)	Lem. 1.1
$\pi_{\mathfrak{p}}^{\widehat{\mathbb{Z}}}(\text{Frobenius})$	λ_i (longitude)	Lem. 1.1
$\mathbb{I}_S$	$H_1(\partial X; \widehat{\mathbb{Z}})$	Thm. 1.4
$\overline{K_{S,+}^\times}$	$\ker \iota_* = \langle \widetilde{\lambda}_i \rangle$	Thm. 1.3, 1.4
$\mathcal{O}_{K,+}^\times$	$\ker \iota_* \cap \bigoplus \widehat{\mathbb{Z}}\mu_i; = 0$ for S^3	Thm. 1.4
$\text{Cl}_S^+ = \langle [\mathfrak{p}] \rangle \leq \text{Cl}_K^+$	$\text{Cl}_{\mathcal{L}} = \langle [K_i] \rangle \leq H_1(M; \mathbb{Z})$	Thm. 1.4, 4.1
$h_S^+ > 1$ obstruction	$[K_i]$ summably in a proper $H < \text{Cl}_{\mathcal{L}}$	Thm. 4.1(2)
—	$H_1(M; \mathbb{Z})/\text{Cl}_{\mathcal{L}} \neq 0$: broken at all β	Thm. 4.1(1)
$\sigma_{\mathfrak{p}} = (\pi_{\mathfrak{p}})_{\mathfrak{q} \neq \mathfrak{p}}$	$\lambda_i = \sum_j \text{Lk}(K_i, K_j)\mu_j$	Prop. 6.2, Thm. 3.2
product formula	linking relation	Rem. 6.3
$\zeta_{K,S}(\beta)$	$\sum_i N(K_i)^{-\beta}$, N geometric on generators	Def. 2.2, 5.1
Ξ_β Dirichlet criterion	Ξ_β linking criterion	Thm. 3.1, 3.2
Chebotarev density	bit matrix: each $\chi \neq 1$ detected, density $\geq \frac{1}{2}$	Thm. 3.5
BC phase diagram	realized by linking, open or closed	Thm. 3.5, Prop. 5.4

The transfer is complete at the level of structure and, by Proposition 2.5, requires no new operator algebra: the classification of [7] is a theorem about a quadruple with no arithmetic in it. What the topological instantiation contributes is a different mechanism for the same phenomenon — linking in place of reciprocity — and §6 shows the two are structurally identical, both being ways of spreading a local generator over infinitely many places.

Three questions. First, realization: [8, Thm. 5.1] realizes prescribed chains $\{\Xi_\beta\}$ from congruence conditions on primes, using Chebotarev to select primes in prescribed classes; Lemma 3.4 removes every constraint on the linking matrix, so the topological realization problem is purely combinatorial, and Theorem 3.5 settles the extreme case. Second, geometry: Remark 5.5 records that we have no family which is simultaneously linking-rich and geometric in its norm, and this is where a genuinely three-dimensional input would enter. Third, and we state it as a caution rather than a programme: nothing here uses a quantum invariant. The Jones polynomial, Chern–Simons theory and the volume conjecture play no role in any proof above, and the subfactors of Paper II of the series arise from arithmetic symmetry breaking rather than from knot theory. We see no

evidence that the two occurrences of subfactors are related, and would not want the coincidence of names to suggest otherwise.

REFERENCES

- [1] M. Kapranov, A. Smirnov, *Cohomology determinants and reciprocity laws*, preprint, 1995.
- [2] B. Mazur, *Remarks on the Alexander polynomial*, unpublished notes, 1963–64.
- [3] M. Morishita, *Knots and Primes: An Introduction to Arithmetic Topology*, Universitext, Springer, 2012.
- [4] J. Neukirch, *Algebraic Number Theory*, Grundlehren 322, Springer, 1999.
- [5] A. Reznikov, *Three-manifolds class field theory*, Selecta Math. (N.S.) 3 (1997), 361–399.
- [6] W. P. Thurston, *The Geometry and Topology of Three-Manifolds*, Princeton lecture notes, 1980.
- [7] R. Chen, *KMS state uniqueness and phase transitions in localized Bost–Connes systems: from Chebotarev density to the Hardy–Littlewood conjecture*, preprint, 2026, DOI 10.5281/zenodo.22152101.
- [8] R. Chen, *Localized Bost–Connes systems I: factor type and the spectrum of transitions*, preprint, 2026, DOI 10.5281/zenodo.22160827.

GUT GEOSERVICE INC., MONTREAL, CANADA
Email address: ruqing@hotmail.com